

CALIBRATION DATA SHEET

Customer Name:

APO CEMENT CORPORATION

Address Line 1:

National Highway, City of Naga, Cebu

Address Line 2:

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Date of Calibration:

Monday, 1 June 2026

Mode:

Compression

Request Ref. No.:

11-2025-FORC-0272(1)

Capacity:

600

Calibration Item:

Compression Testing Machine

Range:

1 kN to 500 kN

Load Cell

Make / Model No.: Controls/C92P02

Serial No. / ID No.: 20006222

Increment:

0.01 kN

Indicator

Make / Model No.: 65-C93P02

Serial No. / ID No.: N/A

Resolution:

0.01 kN

2.) PRE-LOADING DATA, kgf

Indicated force	1st (0°)	Indicated force	2nd (120°)	Indicated force	3rd (240°)
0.00	0.00000	0.00	0.00000	0.00	0.00000
0.00	0.00000	0.00	0.00000	0.00	0.00000
0.00	0.00000	0.00	0.00000	0.00	0.00000

Conversion Unit:
1 kgf = 0.0098067 kN

2.) MEASURED DATA, kgf

Calibration Points	Expected Machine Indication	Indicated force	1st	Indicated force	2nd	Indicated force	3rd	Mean Force	Mean deflection
0	0.00	0.00	0.01399	0.00	0.01399	0.00	0.01399	0.00	0.01399
1st	75.00	74.95	0.06849	74.83	0.06853	74.90	0.06858	75.14	0.06853
2nd	150.00	149.93	0.12270	149.93	0.12283	149.88	0.12292	149.73	0.12282
3rd	200.00	199.93	0.15881	199.92	0.15900	199.88	0.15907	199.39	0.15896
4th	250.00	249.92	0.19508	249.88	0.19530	249.80	0.19538	249.28	0.19525
5th	300.00	299.72	0.23150	299.91	0.23177	299.92	0.23181	299.32	0.23169
6th	350.00	349.87	0.26811	349.92	0.26832	349.88	0.26834	349.46	0.26826
7th	400.00	399.92	0.30482	399.93	0.30504	399.90	0.30497	399.76	0.30494
8th	450.00	449.95	0.34167	449.93	0.34175	449.88	0.34165	450.15	0.34169
9th	500.00	500.00	0.00000	499.92	0.37849	499.98	0.37850	327.31	0.25233
10th	--	--	0.00000	--	0.00000	--	0.00000	0.00	0.00000
0	0	0.00	0.00000	0.00	0.00000	0.00	0.00000		

Max Zero =

3.) Force Transducer Information

In: Compression Mode			
Load Cell	WBRC-R2M	A	1.376099E+03
Capacity	2 MN	B	-9.575582E+00
Serial No.	WB-121005	C	4.373055E+00

4.) Environmental Conditions:

Thermo-hygrometer indication		
	Before	After
Temp:	23.9	25.0
Humidity:	55.0	54.0

Cal. Cert. No.	2400-00342-003	U, %	0.08
Cal Date	May 24, 2024	k	2.00

Calibrated by:

RFM

ACCG

Checked by:

date:

- ☐ no corrections
- ☐ with corrections

5.) Net value (dij) & Mean Value (di) Calculation, units in kN

Calibration Points	Force (kN)	1st (0°)	Force (kN)	2nd (120°)	Force (kN)	3rd (240°)	Mean Force (kN)	Mean deflection
0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
1st	74.95000	0.05450	74.83000	0.05454	74.90000	0.05459	75.13612	0.05462
2nd	149.93000	0.10871	149.93000	0.10884	149.88000	0.10893	149.73497	0.10889
3rd	199.93000	0.14482	199.92000	0.14501	199.88000	0.14508	199.39482	0.14504
4th	249.92000	0.18109	249.88000	0.18131	249.80000	0.18139	249.28057	0.18136
5th	299.72000	0.21751	299.91000	0.21778	299.92000	0.21782	299.32208	0.21781
6th	349.87000	0.25412	349.92000	0.25433	349.88000	0.25435	349.45859	0.25435
7th	399.92000	0.29083	399.93000	0.29105	399.90000	0.29098	399.76085	0.29101
8th	449.95000	0.32768	449.93000	0.32776	449.88000	0.32766	450.15314	0.32776
9th	500.00000	-0.01399	499.92000	0.36450	499.98000	0.36451	327.30515	0.23836
10th	0.00000	-0.01399	0.00000	-0.01399	0.00000	-0.01399	0.00000	0.00000
0	0.00000	-0.00014	0.00000	-0.00014	0.00000	-0.00014		

6.) Linear Interpolation

Force (kN)	Load Value			
	1st (0°)	2nd (120°)	3rd (240°)	Mean Value
0.0000	0.00000	0.00000	0.00000	0.00000
75.0000	0.05454	0.05466	0.05466	0.05462
150.0000	0.10876	0.10889	0.10902	0.10889
200.0000	0.14487	0.14507	0.14517	0.14504
250.0000	0.18115	0.18140	0.18154	0.18136
300.0000	0.21771	0.21785	0.21788	0.21781
350.0000	0.25421	0.25439	0.25444	0.25435
400.0000	0.29089	0.29110	0.29105	0.29101
450.0000	0.32772	0.32781	0.32775	0.32776
500.0000	-0.01399	0.36456	0.36452	0.23836
0.0000	--	--	--	0.00000
0.0000	0.00000	0.00000	0.00000	0.00000

7. Estimation of Reference Force

Force (kN)	Load Value			
	1st (0°)	2nd (120°)	3rd (240°)	Mean Value
0.0000	0.00000	0.00000	0.00000	0.00000
75.0000	75.01966	75.19505	75.19364	75.13612
150.0000	149.55792	149.73665	149.91034	149.73497
200.0000	199.16876	199.43980	199.57589	199.39482
250.0000	248.98931	249.33135	249.52104	249.28057
300.0000	299.18617	299.36756	299.41251	299.32208
350.0000	349.27723	349.51560	349.58295	349.45859
400.0000	399.58832	399.88016	399.81407	399.76085
450.0000	450.09574	450.22544	450.13823	450.15314
500.0000	-19.25351	500.60761	500.56135	327.30515
0.0000	--	--	--	0.00000
0.0000	0.00000	0.00000	0.00000	0.00000

8.) Uncertainty Calculations

Relative Expanded	Calibration Points	Relative uncertainty	Rel unc due to res. (w _{res})	Relative std uncertainty	Relative combined	Relative expanded	Relative accuracy error a	Relative repeatability error b	Relative zero error f ₀	Class
8.00E-02	0	--	--	--	--	--	0.00	0.00	0.00	Outside Class
Resolution	1st	0.077500	0.003842	0.049329	0.092112	0.184	-0.18	0.23	0.00	Class 2
0.4903325	2nd	0.067945	0.001928	0.049329	0.084165	0.168	0.18	0.24	0.00	Class 2
temp sens.	3rd	0.060012	0.001448	0.049329	0.077892	0.156	0.30	0.20	0.00	Class 2
0.003	4th	0.062413	0.001158	0.049329	0.079751	0.160	0.29	0.21	0.00	Class 2
Temp Diff.	5th	0.023113	0.000964	0.049329	0.054761	0.110	0.23	0.08	0.00	Class 2
1.1	6th	0.026538	0.000826	0.049329	0.056290	0.113	0.15	0.09	0.00	Class 2
Drift Sens.	7th	0.022100	0.000722	0.049329	0.054337	0.109	0.06	0.07	0.00	Class 2
0.05	8th	0.008480	0.000641	0.049329	0.050358	0.101	-0.03	0.03	0.00	Class 2
	9th	52.941217	0.000882	0.049329	52.941241	105.883	52.76	158.83	0.00	Outside Class
	10th	--	--	--	--	--	0.00	0.00	0.00	Outside Class

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